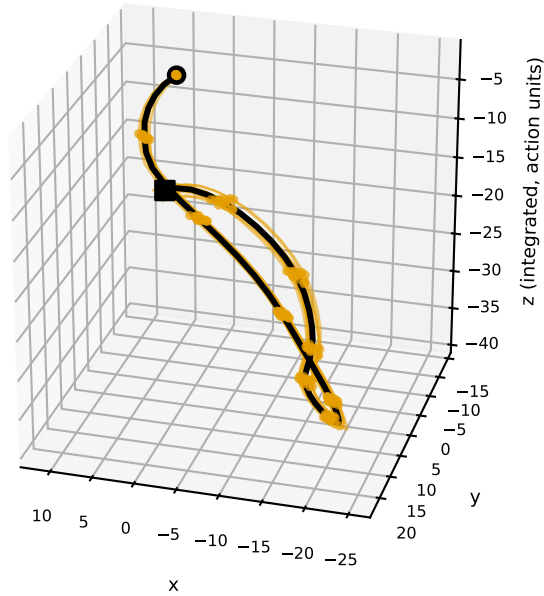


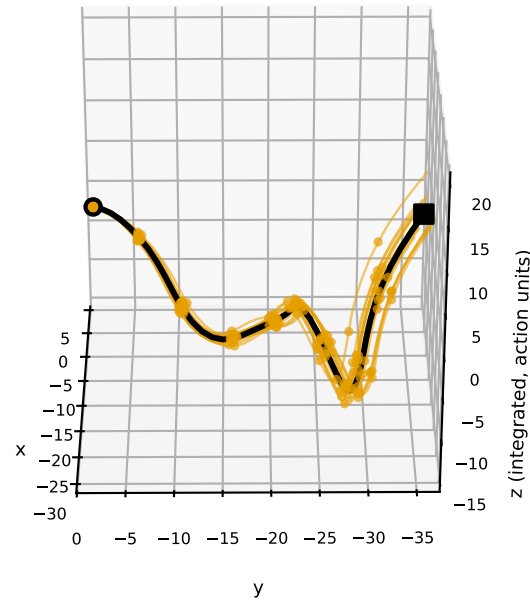
Sampled action chunks as 3-D paths (thin: each sample; black: mean, o start / ■ end; dots every 5 steps for $\pi_{0.5}$, every step for GR1).

Top: $\pi_{0.5}$ end-effector translation x,y,z. Bottom: GR1, offsets of the three arm joints with the largest motion (normalized units). Equal scale on all three axes; camera tilted 25-30° off the plane of the mean path.

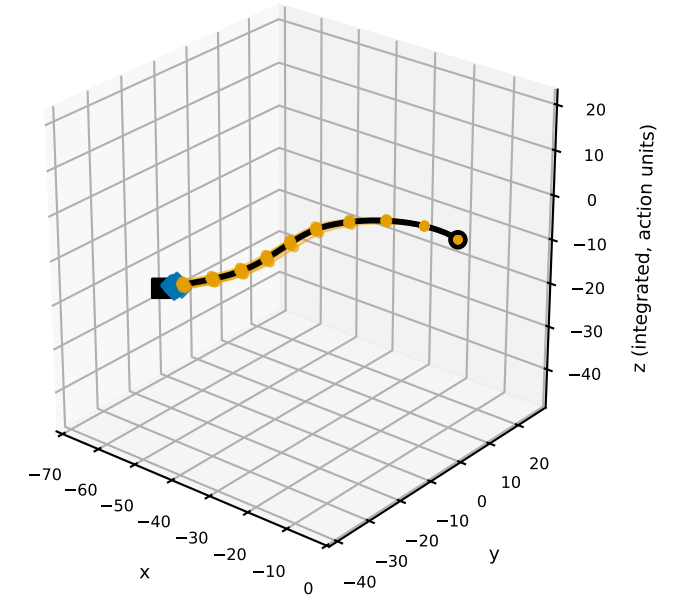
$\pi_{0.5}$ flow, typical state (sep 3.6)
path length 90.8; tube width mean 0.57 / max 1.98



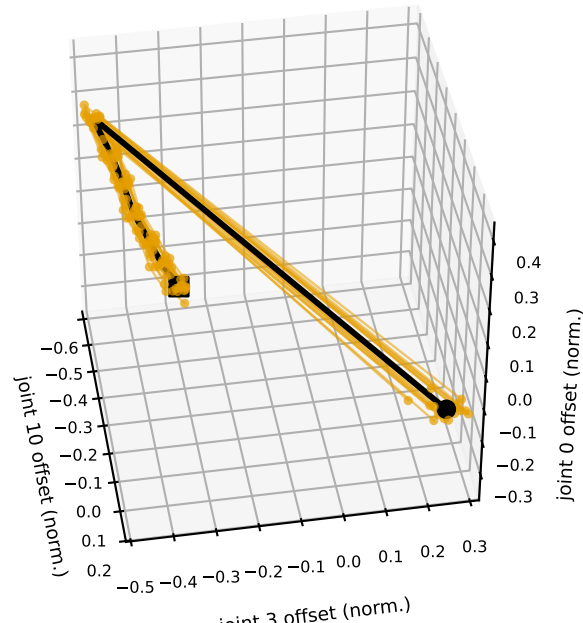
$\pi_{0.5}$ flow, widest state (sep 0.0)
path length 69.2; tube width mean 0.87 / max 4.78



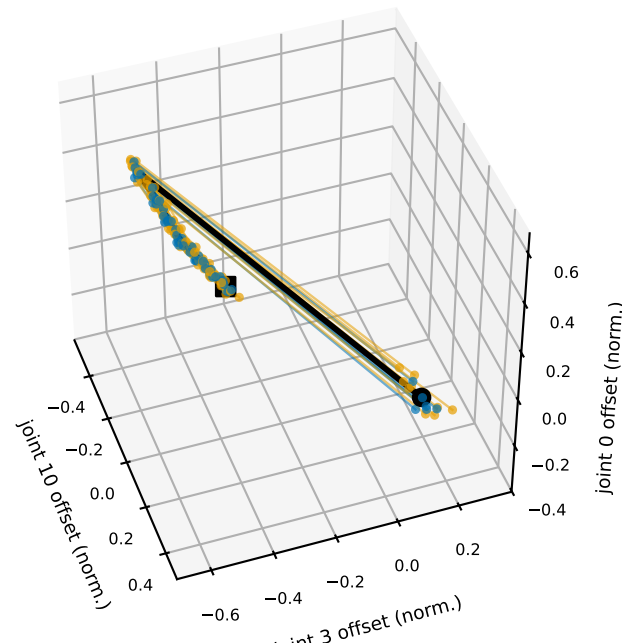
$\pi_{0.5}$ flow, most separated state in 3000 (sep 257, K=32)
orange = gripper open, blue = closed (◆ = close step); step 46 in 22 samples, step 47 in 10 samples
path length 76.9; tube width mean 0.43 / max 1.30



GR1 flow, typical state (sep 2.5)
all samples give the same hand command (channel 18)
path length 1.6; tube width mean 0.03 / max 0.09



GR1 flow, widest state (sep 0.0)
hand closes inside the chunk (channel 24)
orange: by step 4 (n=9), blue: later (n=7)
path length 1.6; tube width mean 0.04 / max 0.14



GR1 flow, most separated state in 3000 (sep 8.7, K=32)
hand channel 19 at two levels all chunk long
blue: high (n=18), orange: low (n=14)
path length 1.4; tube width mean 0.11 / max 0.23

